THE MASTERMIND COACHING CLASSES

CBSE / RBSE • NURSERY – CLASS XII • JAIPUR

Road No. 5, VKIA, Jaipur — 9887258679

CLASS 10 — MATHEMATICS FULL SYLLABUS PAPER SET 05

Full Syllabus (Chapters 1–14) | RBSE Board Pattern | Session 2026–27

Time: 3 Hours 15 Minutes Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–28 are Very Short Answer Type (2 marks each).
4. Q. 29–36 are Short Answer Type (3 marks each).
5. Q. 37–40 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 35 to 40.
7. Write the question number before each answer.
8. Solve Q. 29 on graph paper. Draw neat figures.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. The prime factorisation of 432 is: 1

(a) $2^2 \times 3^3$ **(b)** $2^3 \times 3^4$ **(c)** $2^4 \times 3^2$ **(d)** $2^4 \times 3^3$

2. The HCF of $2^4 \times 3$ and $2^2 \times 3^2 \times 5$ is: 1

(a) 720 **(b)** 6 **(c)** 12 **(d)** 4

3. If the zeroes of $ax^2 + bx + c$ are reciprocals of each other, then: 1

(a) $a = b$ **(b)** $b = c$ **(c)** $a = -c$ **(d)** $a = c$

4. The zeroes of $x^2 - 1$ are: 1

(a) no real zeroes **(b)** ± 1 **(c)** 0, 1 **(d)** 1 only

5. The pair $3x + 4y = 5$ and $6x + 8y = 7$ has: 1

(a) a unique solution **(b)** infinitely many solutions
(c) two solutions **(d)** no solution

6. If $x = 1,\ y = 1$ is a solution of $9ax + 12ay = 63$, then a is: 1

(a) 7 **(b)** 3 **(c)** 21 **(d)** 1

7. "The product of two consecutive odd integers is 63" is represented by: 1

(a) $x^2 - 63 = 0$ **(b)** $x^2 + x - 63 = 0$ **(c)** $x^2 + 2x - 63 = 0$ **(d)** $2x + 2 = 63$

8. The discriminant of $\sqrt{3}x^2 + 2\sqrt{2}x - 2\sqrt{3} = 0$ is: 1

(a) 32 **(b)** 16 **(c)** −16 **(d)** 8

9. For the number of multiples of 4 between 10 and 250, Student A says 60; Student B says 61. Then: 1

(a) Student A is correct **(b)** Student B is correct
(c) Both are correct **(d)** Both are wrong

10. The 6th term of the AP with $a_n = 4n - 3$ is: 1

(a) 24 **(b)** 21 **(c)** 25 **(d)** 17

11. A student states: "If two angles of one triangle equal two angles of another, the triangles are congruent." The correct response is: 1

(a) Correct
(b) Correct for equilateral triangles
(c) Incorrect — they are similar, not necessarily congruent
(d) Cannot be determined

12. A student states: "Two triangles with equal areas are similar." The correct response is: 1

(a) Correct
(b) Incorrect — equal area does not imply equal angles
(c) Correct for right triangles
(d) Cannot be determined

13. $\sin^2 A + \frac{1}{1+\tan^2 A}$ equals: 1

(a) $\sec^2 A$ **(b)** 0 **(c)** 1 **(d)** $2\sin^2 A$

14. A student states: "The tangent at a point of a circle is perpendicular to every chord through that point." The correct response is: 1

(a) Correct
(b) Correct for the diameter only
(c) Incorrect — it is perpendicular to the radius through that point
(d) Cannot be determined

15. The area of a sector of radius 14 cm and angle 90° is: 1

(a) 154 cm² **(b)** 308 cm² **(c)** 77 cm² **(d)** 616 cm²

16. The area swept by a 14 cm minute hand in 30 minutes is: 1

(a) 154 cm² **(b)** 44 cm² **(c)** 616 cm² **(d)** 308 cm²

17. The volume of a hemisphere of radius 7 cm is: 1

(a) 718.67 cm³ **(b)** 359.33 cm³ **(c)** 308 cm³ **(d)** 1437.33 cm³

18. The class width of the classes 10–20, 20–30, … is: 1

(a) 5 **(b)** 15 **(c)** 10 **(d)** 20

19. A student states: "The step-deviation method gives a different mean from the direct method." The correct response is: 1

(a) Correct **(b)** Incorrect — both give the same mean
(c) It is always larger **(d)** Cannot be determined

20. Two coins are tossed. The probability of two tails is: 1

(a) $\frac{3}{4}$ **(b)** 1 **(c)** $\frac{1}{2}$ **(d)** $\frac{1}{4}$

SECTION — B (Very Short Answer Type) 8 × 2 = 16

Answer in about 20–30 words each.

21. Given that $\text{LCM}(306, 657) = 22338$, find $\text{HCF}(306, 657)$. 2

22. Find the zeroes of $3x^2 - 75$ and verify the relationship between the zeroes and the coefficients. 2

23. Find the nature of the roots of $2x^2 - 3x + 5 = 0$. 2

24. In $\triangle ABC$, $DE \parallel BC$ with $AD = 6$ cm, $DB = 9$ cm and $AE = 8$ cm. Find AC. 2

25. If M(5, −1) is the mid-point of AB and A is (2, 3), find B. 2

26. Evaluate: $\frac{\cos 45^\circ}{\sec 30^\circ + \csc 30^\circ}$. 2

27. The sun's elevation is 60° and a tower's shadow is 10 m. Find its height. 2

28. The incircle of $\triangle ABC$ touches AB at F, BC at D and CA at E. If $AF = 4$ cm, $BD = 3$ cm and $CE = 5$ cm, find the perimeter of the triangle. 2

SECTION — C (Short Answer Type) — 8 × 3 = 24

Answer in about 40–60 words each. Solve Q. 29 on graph paper.

29. Draw the graphs of the following pair of linear equations on graph paper. Determine whether the pair is consistent; if so, find the solution from the graph: $2x - y = 6$ and $x + y = 0$. 3

30. Find the median of: 3

Class	10–20	20–30	30–40	40–50	50–60
Frequency	5	9	14	8	4

31. The 8th term of an AP is 31 and its 15th term is 16 more than its 11th term. Find the AP. 3

32. Prove that the tangent drawn at the mid-point of an arc of a circle is parallel to the chord joining the end points of the arc. 3

33. A circular park of radius 21 m has a sector-shaped pond of angle 60°. Find the area of the pond and the remaining area of the park. 3

34. A bag has 5 white and 7 red balls. Find P(red). If one red ball is removed, find the new P(red). 3

35. From the top of a 30 m tower, the angles of depression of two points on the ground on the same side are 30° and 60°. Find the distance between the points. 3

OR

An observer 1.5 m tall is 28.5 m from a 30 m tower. Find the angle of elevation of the top from his eyes. How far should he be for the angle to be 60°? 3

36. Find the median life time of 400 lamps: 3

Life (hrs)	1500–2000	2000–2500	2500–3000	3000–3500	3500–4000	4000–4500	4500–5000
Lamps	14	56	60	86	74	62	48

OR

Find the mean literacy rate by the step-deviation method: 3

Literacy (%)	45–55	55–65	65–75	75–85	85–95
Cities	3	10	11	8	3

SECTION — D (Long Answer Type) — 4 × 5 = 20

Answer in about 100–150 words each.

37. Show that $(7, 10)$, $(-2, 5)$ and $(3, -4)$ are the vertices of an isosceles right triangle. Show that the mid-point of the hypotenuse is equidistant from all three vertices. 5

OR

If P$(9a - 2, -b)$ divides the join of A$(3a + 1, -3)$ and B$(8a, 5)$ in the ratio 3 : 1, find a and b. Then find AB. 5

38. Prove that $\frac{\sin A - 2\sin^3 A}{2\cos^3 A - \cos A} = \tan A$. Verify it for $A = 30^\circ$. 5

OR

Prove that $\frac{1 - \sin\theta}{1 + \sin\theta} = (\sec\theta - \tan\theta)^2$. 5

39. A conical cavity of radius 7 cm and height 24 cm is hollowed out from a solid cylinder of the same radius and height. Find 5

(i) the volume of the remaining solid

(ii) its total surface area.

OR

A boiler is a cylinder 2 m long with hemispherical ends, its diameter being 1 m. Find 5

(i) its volume

(ii) its surface area.

40. Find the mean and median of the runs scored: 5

Runs	0–20	20–40	40–60	60–80	80–100
Players	7	10	15	12	6

OR

Case study: Marks of 60 students are given below. Find 5

(i) the mean

(ii) the median

(iii) the mode using the empirical relationship.

Marks	0–10	10–20	20–30	30–40	40–50	50–60
Students	4	8	12	20	10	6

END OF QUESTION PAPER **Total: 80 Marks**

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